**“Let’s not ask our brains to be in two places at once. During lectures and class discussions, close your laptop and be fully present – we will open it again when it is time to code.”**

# PRESEMO Feedback

## I like …

13:58 (21.10) » Introduction to different techniques for evaluating regression models. ☆

13:58 (21.10) » Gragh in the slide ☆

13:58 (21.10) » Good variety of indicators ☆

## I wish …

13:58 (21.10) » More real example during lecture ☆

13:58 (21.10) » Good ☆

13:58 (21.10) » Use of references in slides ☆

# Quiz Feedback

- All learning objectives were met – mostly "independently" and a few "with some support"
- There was one "not there yet" for 2/3 learning objectives, please contact me or the TAs for a one-on-one support session if still needed

# Where we are on the course

We go at a pace that makes sense and adjust the depth/breadth of topics

## Period I

- L1: Course orientation and foundations of marine data analysis
- L2: Exploratory Data Analysis (EDA)
- L3 & L4: Data pre-processing
- L5: From engineering questions to analytical models
- L6: Marine data analysis in practice

## Period II

- L7: Regression modelling, evaluation, and refinement
- **L8 & L9: AIS data analytics**
- L10 & L11: Cluster analysis
- L12: AI-assisted qualitative data analysis
- Project demonstration

Aalto-yliopisto
Insinööritieteiden
korkeakoulu

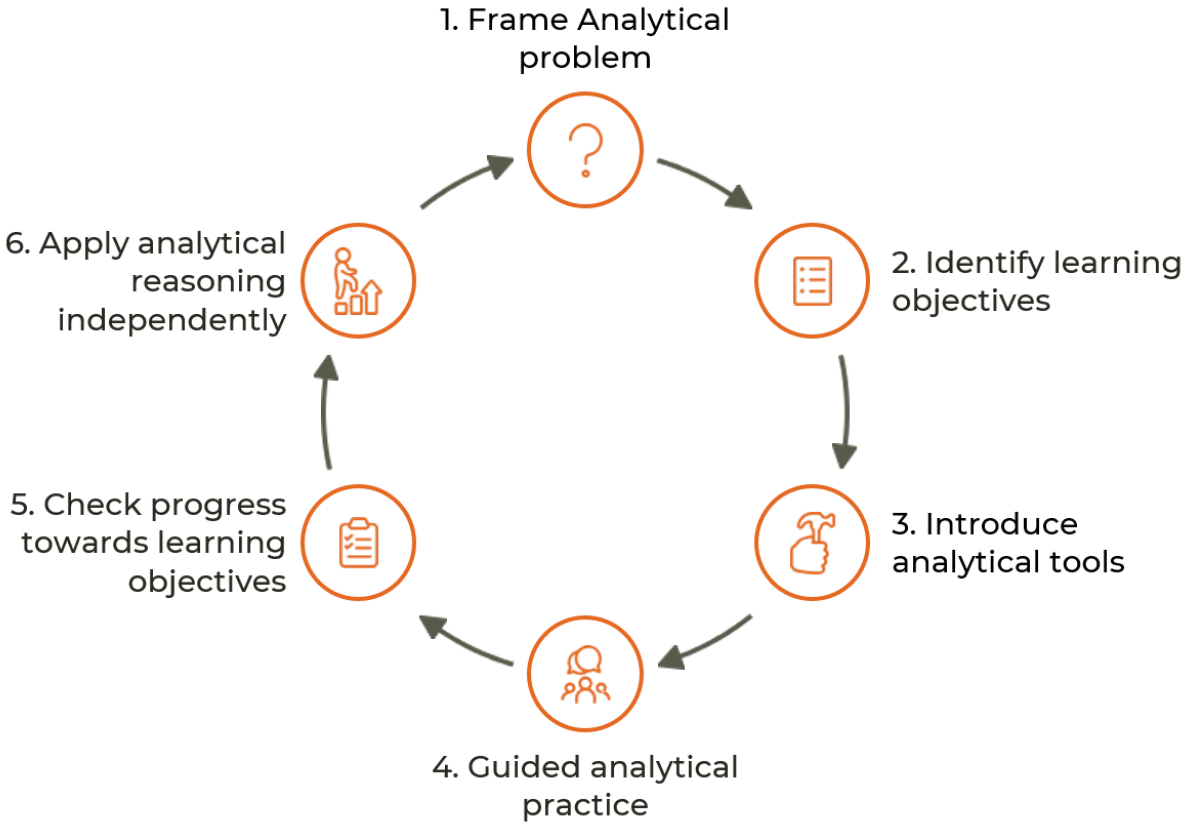
# Introduction to AIS and AIS data analytics

MEC-E2016 Marine Data Analysis

Reference: Contents of this lecture are partially covered in

1. Guide to Maritime Informatics. Chapter 1 (Section 1.2), Chapter 4 (Sections 4.1 and 4.2).

# Problem based learning cycle



# Problem of the week

**Grade 1 to 3: AIS data preprocessing**

Apply appropriate preprocessing techniques to the AIS dataset, *AIS_data_timestamp.csv*, included in AIS_Preparation.zip. The ZIP file also contains a *libs* folder with Python functions for preprocessing and map visualization, and a *README.md* file explaining their use.

In your report, describe and justify each preprocessing action and explain its effect on data quality. Visualize the vessel positions before and after cleaning, compare the results, and discuss any remaining limitations that could affect further analysis.

**Grade 4 to 5: Evaluation of AI-assisted preprocessing**

In addition to the above, repeat the AIS preprocessing task using a generative AI tool of your choice. State which tool you used and briefly describe how you instructed it. Compare its approach and results with your own work, focusing on the preprocessing decisions and their justification. Explain which AI suggestions you would accept, modify or reject, and support your assessment with specific examples from the dataset.

# Learning objectives

At the end of this lecture, you should be able to

- Describe Automatic Identification System (AIS) – equipment, attributes in AIS messages
- Describe unique challenges and difficulties of using AIS data

# Automatic Identification System (AIS)

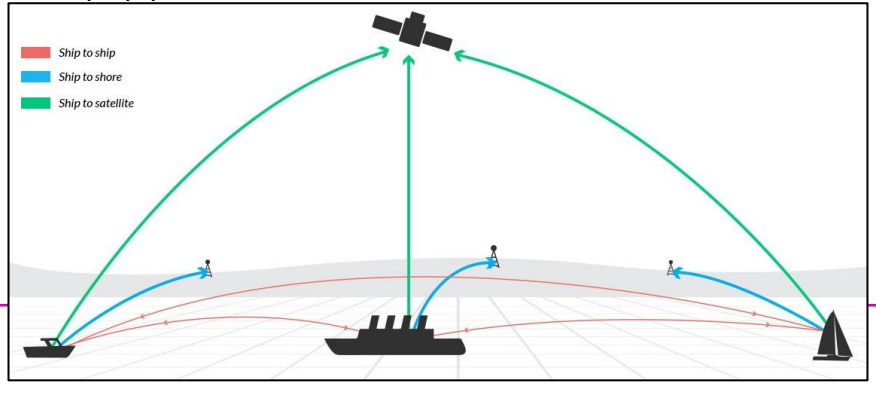
- AIS – An automatic tracking system that uses transceivers on ships and is used by vessel traffic services(VTS)

- International Maritime Organization (IMO) mandates AIS transceivers installation on-board of many commercial vessels

- The system automatically broadcasts navigational information about vessels along with vessel characteristics (e.g., name) to all other installations (e.g., AIS stations, on-board transceivers) in a self-organized manner

# AIS Equipment

- AIS is based on the use of dedicated equipment that should be installed
  - aboard (vessel stations)
  - ashore (base stations)
  - on dedicated satellites (AIS-SAT)
- Vessels are equipped with transponders, e.g., stations that send and receive AIS messages
- Base stations that are installed ashore are equipped with AIS receivers (some also have transmitters)
- Dedicated satellites are also equipped with AIS receivers



# AIS Data

- AIS data refers to information collected through the Automatic Identification System (AIS)
- Primary use: safety of navigation in ship-to-ship use (e.g., prevention of collisions)
- But now used also for
  - Monitoring vessels
  - extracting valuable information regarding vessel behavior
  - Identifying operational pattern
  - Calculating performance statistics
  - And many more issues

# AIS Messages

- Three categories: **dynamic**, **static**, and **voyage-related** messages along with ship identity number (**MMSI or IMO** number)

- Some attributes in the messages use direct values while others are encoded.

# AIS Messages – Dynamic message

- **Dynamic** - This is real-time information that is updated frequently, providing a snapshot of the vessel's current status

| Attribute | Description |
|---|---|
| Ship position | Latitude and longitude |
| Speed over ground | Ranging from 0 knots to 102 knots (0.1knot resolution) |
| Course over ground | The direction shows the actual direction of movement over the ground(0° to 359° ) |
| Heading | The direction the ship's bow is pointing (0° to 359° ); |
| Rate of Turn (ROT) | It indicates how fast a vessel is turning (°/minute) (from -126 to 126);<br>A positive value indicates a turn to the right (starboard), while a negative value indicates a turn to the left (port);<br>127, and -127 indicate no ROT available;<br>0 indicates not turning. |
| Timestamp | The time that tells when the dynamic information (position, COG, SOG, ROT, heading, etc.) was generated |
| Navigation status | Includes at anchor\under way using engine(s) \not under command\others |

| timestamp | MMSI | Latitude | Longitude | Heading | Course_over_groud | Speed_over_ground | ROT | Navigational_status |
|---|---|---|---|---|---|---|---|---|
| 1/15/2020 2:00:00 | 230011000 | 63.01936 | 20.54013 | 195 | 197 | 6.9 | 0 | Under way using engine |
| 1/15/2020 2:00:00 | 265522440 | 59.32846 | 18.07654 | 344 | 0 | 0 | 0 | Moored |
| 1/15/2020 2:00:00 | 265412000 | 62.40839 | 17.39126 | 146 | 15.8 | 0 | 0 | At anchor |
| 1/15/2020 2:00:00 | 230938490 | 58.94188 | 17.97525 | 145 | 4.2 | 0 | 0 | At anchor |
| 1/15/2020 2:00:00 | 566577000 | 58.00303 | 20.3865 | 24 | 24.1 | 12.4 | -1.61 | Under way using engine |

Dynamic message examples

# AIS Messages – Static message

- **Static** - This data is fixed and only needs to be updated when a major change occurs, such as a change in the vessel's identity or dimensions

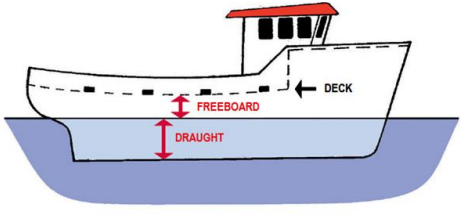
| Data field | Description |
|---|---|
| Ship identity | Maritime Mobile Service ID (MMSI) (9-digit ID) and IMO number. |
| Call sign | The international radio call sign assigned to the vessel by her country of registry |
| Ship name | Name of the ship |
| Ship type | Cargo, tanker, passenger, etc. The ship type is shown as a numeric code because AIS encodes ship type using standardized numbers |
| Ship dimension | Dimensions of the ship in meters. Length and beam(width) |
| Reference to position of antenna | The location of the AIS GPS antenna relative to the ship's hull |

| Timestamp | MMSI | IMO | Name | Call_sign | Breath | Length | Type |
|---|---|---|---|---|---|---|---|
| 1/15/2010 0:00 | 273310710 | 9344875 | ASCOLD | UCSZ | 0 | 0 | 52 |
| 1/15/2010 0:00 | 276784000 | 0 | PILOT AHTO-01 | ES2819 | 4 | 18 | 50 |
| 1/15/2010 0:00 | 266022000 | 9256432 | ALCEDO | SBHD | 18 | 118 | 80 |
| 1/15/2010 0:00 | 276421000 | 9222417 | MARS | ESPS | 11 | 30 | 52 |
| 1/15/2010 0:00 | 230286000 | 7927958 | ESKO | OIOI | 10 | 30 | 52 |

# AIS Messages – Voyage-related message

- **Voyage-related -** This data describes the ship's journey and is typically entered by the crew and updated as needed

| Data field | Description |
|---|---|
| Destination | Destination port |
| Estimated time of arrival (ETA) | The estimated time of arrival of the ship |
| Draught/Draft | The vertical distance between the waterline and the bottom of the hull (keel), with the thickness of the hull included |




| Timestamp | MMSI | Destinatic | ETA_UTC | Draft_m |
|---|---|---|---|---|
| 9/2/2025 7:36 | 636234055 | ABU DHAB | 10/24/2025 8:40 | 13.6 |
| 8/17/2025 2:37 | 538590457 | DURBAN | 9/29/2025 23:19 | 4.8 |
| 9/3/2025 3:10 | 273776022 | SALALAH | 10/5/2025 20:09 | 9.2 |
| 9/15/2025 7:23 | 370909942 | DURBAN | 10/4/2025 11:22 | 6.8 |
| 8/23/2025 4:08 | 374833547 | SALALAH | 9/26/2025 13:04 | 12.5 |

# AIS Messages – Why attribute set varies?

**Table 1**. Attributes of AIS messages exchanged using Class A and Class B transponders

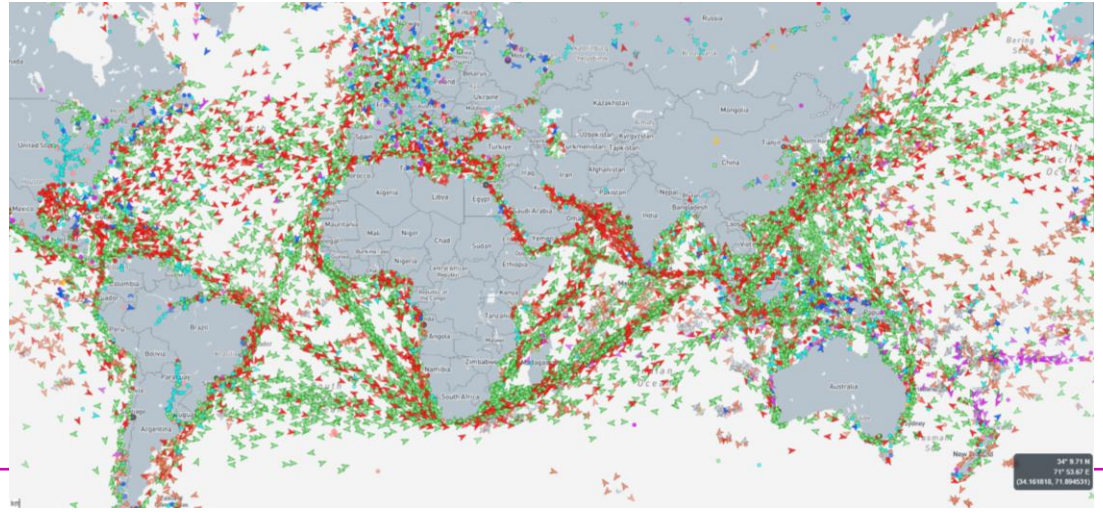
| Data | Class A (receive) | Class B (send) | Class B (receive) |
|---|---|---|---|
| Call sign | Yes | Yes | Yes |
| IMO | Yes | No | No |
| Length and beam | Yes | Yes | Yes |
| Antenna location | Yes | Yes | Yes |
| Draught | Yes | No | No |
| Cargo information | Yes | Yes | Yes |
| Destination | Yes | No | No |
| Estimated time of arrival | Yes | No | No |
| Time | Yes | Yes | Yes |
| Ship's position | Yes | Yes | Yes |
| Course over ground | Yes | Yes | Yes |
| Speed over ground | Yes | Yes | Yes |
| Gyro heading | Yes | Yes | Yes |
| Rate of turn | Yes | No | No |
| Navigational status | Yes | No | No |
| Safety message | Yes | No | Yes |

- Attribute list may vary depending on
  - Type of Transponders (Class A vs. Class B)
  - Type of exchange (Send vs. Receive)

# AIS Processing difficulties and challenges

- Originally, AIS was designed to allow vessels to provide ship information automatically to other ships and to maritime authorities

- Now: AIS is used for vessel tracking services at a global scale
  - Some inherent characteristics of the communication protocol raise technical challenges that should be addressed to offer consistent and reliable information



Snapshot from Marine Traffic

# AIS processing difficulties and challenges

- **Absence of unique ship identification**
  - MMSI is not unique and can change
  - IMO is unique, but not all ships have them
  - The absence of a unique ID requires cross-checking vessel details (e.g., dimensions, name) and assigning an identifier to maintain consistent tracking.
- **Sensor malfunction**
  - A vessel may transmit erroneous information due to sensors' faulty operation
  - This makes feasibility analysis essential to evaluate whether a vessel position is valid based on the vessel's past positions.

# AIS Processing difficulties and challenges

- **Manual input and inconsistent reporting**
  - Some of the information included in AIS messages are manually inserted by the vessel's crew and can contain inconsistencies (e.g., Destination, ETA)
    - A passenger vessel that is performing the same itinerary with multiple stops every day, may not change the destination for each stop and report only the final destination from the beginning of the voyage
    - "Piraeu", "Piraeus Port", "Piraeus Anchorage", "Pir", "P" are all acceptable values in the reported destination of a vessel travelling to the port of Piraeus
    - ETA field may also be unreliable, since crews often report the time to the final destination instead of the next port.
  - This may necessitate extensive pre-processing or other means of estimation
    - For example, calculating the ETA based on the vessel's speed or determining the vessel's destination based on its itinerary history or using pattern-of-life analysis

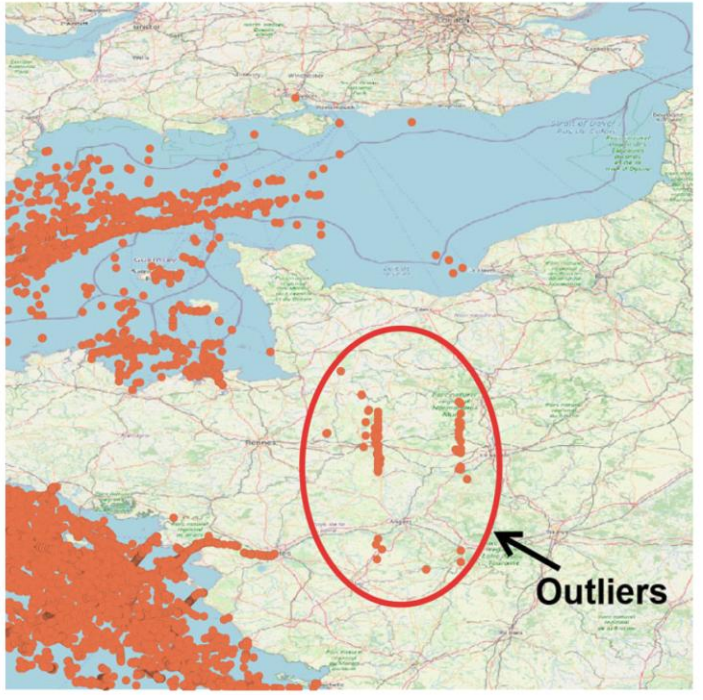
# AIS Processing difficulties and challenges

- **Reporting Frequency**
  - According to the AIS communication protocol, the reporting intervals are fluctuating and depend on the vessel's behavior (e.g., speed, rate of turn).
  - When satellites are used for monitoring, the update interval may range from a few minutes up to several hours, depending on the satellite availability.
  - All these lead to a non-uniform AIS data stream, with communication gaps that cause irregular vessel tracks and missing points between consecutive messages.

| Types of AIS transponders | Condition | Reporting Interval (Dynamic messages) |
|---|---|---|
| Class A & Class B | Anchored/moored, or speed ≤ 2 knots | Every 3 min |
| Class B | 2<Speed ≤ 14 knots | Every 30 s |
| Class B | Speed > 14 knots | Every 15 s |
| Class A | Speed ≤ 14 knots | Every 10 s |
| Class A | Higher speed / turning | Even higher reporting rate (faster than 10 s) |

# AIS Processing difficulties and challenges

- **Geolocation**
  - The geolocation of vessels in messages should be given in a consistent way using a common geo-reference system by all message sources
  - When this is not the case, then locations may not be interpreted correctly, and vessels may appear to be moving on the mainland
  - This issue must be dealt with adequately in a pre-processing step



# AIS Processing difficulties and challenges

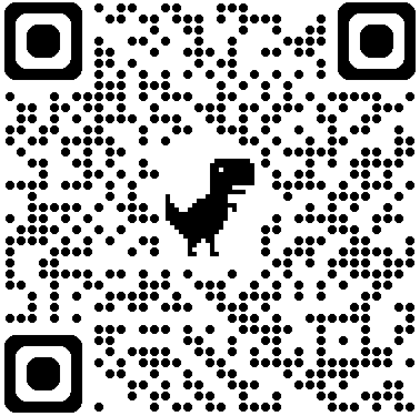
- **Timestamping**
  - AIS position reports commonly include the UTC second within the minute, rather than a complete date and time.
  - The same AIS message may be received by multiple stations, creating duplicates in the data.
  - Messages can reach the central system with variable delays due to processing and network latency.
  - Each record should have a clearly defined timestamp, such as reception time or arrival time at the central system. This can be stored as Unix time: seconds since 1 January 1970, 00:00:00 UTC.

# Exercise for today

- Find a peer and work through the Python exercise together.
- Download **AIS_Preparation.zip** from the assignment 4 description and extract it.
- The extracted folder contains the AIS dataset, a libs folder with Python functions for preprocessing and map visualization, and a README.md file explaining their use.
- We will be working in **AIS_main.ipynb**, which is also included in the ZIP file. **Have this notebook open and ready to run.**
- **Unlike previous weeks, you will attempt the notebook code yourselves using the provided functions and documentation. Work with your peer first, and we will review the code together at the end.**

# Quiz

- Complete your quiz here.



# Feedback

- Provide your feedback here

